Chapter 4 : Quantum Information Science

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In this chapter, we study basic concepts to prepare our journey to the quantum machine learning.

Topics include basic algorithms, quantum parallelism, teleportation, and associated programming in Qiskit and Cirq.

Part 1 : Qiskit and Cirq to construct quantum circuits and states, Von Neumann Entropy

Part 2 : Quantum Teleportation, Gate Scheduling, Quantum Parallelism and Function Evaluation, Deutsch's Algorithm

Part 1 : Quantum Circuits with Qiskit and Cirq , and Von Neumann Entropy

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Quantum states and Observables: A review

A pure state is a vector $|\psi\rangle$ in a Hilbert space $\mathbb{H}$.

In our case, $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$.

An observable A is a self-adjoint operator on $\mathbb{H}$.

We compute the mean value of A by $\langle A\rangle_\psi := \langle\psi|A\psi\rangle$.

The probability $P_\psi(\lambda)$ that for a quantum system in the pure state $|\psi\rangle \in \mathbb{H}$, a measurement of the observable yields the eigenvalue λ of A is given by

$$P_\psi(\lambda) = \|P_\lambda|\psi\rangle\|^2,$$

where P_λ is the projection onto the eigenspace $Eig(A, \lambda)$.

Quantum states and Observables: A review

Projection postulate : If a measurement of the observable A on a quantum mechanical system in the pure state $|\psi\rangle \in \mathbb{H}$ yields the eigenvalue λ, then

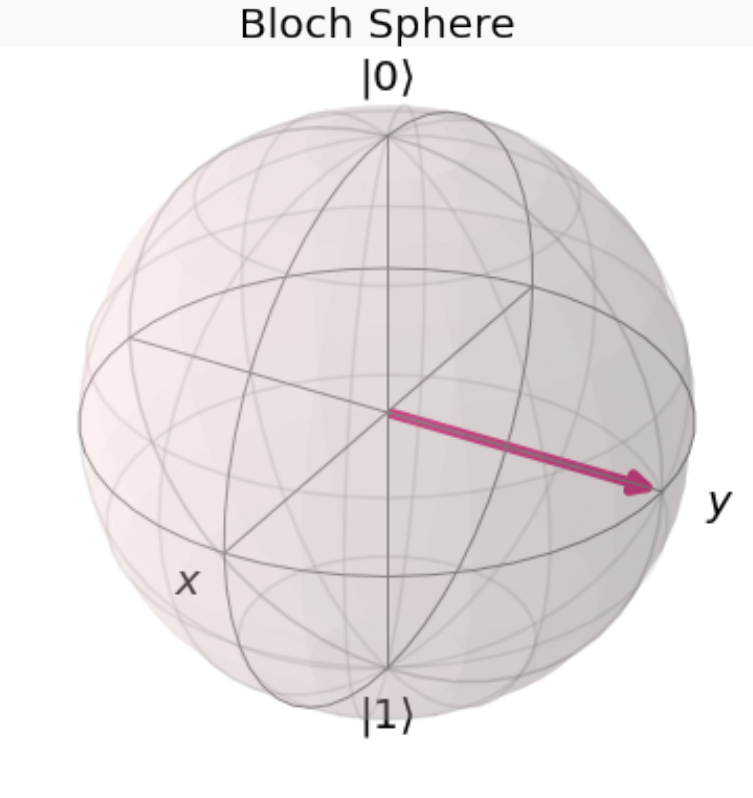
$$|\psi\rangle = \text{state before measurement} \rightarrow \frac{P_\lambda|\psi\rangle}{\|P_\lambda|\psi\rangle\|} = \text{state after measurement.}$$

A density operator is an operator ρ acting on $\mathbb{H}$ which is self-adjoint, postive($\rho \geq 0$), and has trace 1. Then, the quantum states described by density operators are called mixed states.
Indeed, we can write

$$\rho = \sum_j p_j |\psi_j\rangle\langle\psi_j|$$

Bloch representation of pure and mixed states

Since a qubit $|\psi\rangle \in \mathbb{HS}^1$, we can represent it on a sphere.



Bloch representation of pure and mixed states

To be more specific, $|\psi\rangle = a|0\rangle + b|1\rangle$ with $|a|^2 + |b|^2 = 1$ allows a parametrization

$$|\psi\rangle = e^{-\mathrm{i}\frac{\phi}{2}} \cos\frac{\theta}{2}|0\rangle + e^{\mathrm{i}\frac{\phi}{2}} \sin\frac{\theta}{2}|1\rangle.$$

For a density operator

$$\rho = \begin{pmatrix} a & b \\ c & d, \end{pmatrix}$$

using $\rho^* = \rho$ and $tr(\rho) = 1$, we get

$$\rho = \frac{1}{2}(1 + x \cdot \sigma) = \frac{1}{2}\begin{pmatrix} 1 + x_3 & x_1 - \mathrm{i}x_2 \\ x_1 + \mathrm{i}x_2 & 1 - x_3 \end{pmatrix},$$

where $\sigma = (\sigma_x, \sigma_y, \sigma_z)$.

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Getting started with python libraries

Qiskit : Library from IBM

Cirq : Library from Google

Getting started with Qiskit

Good introduction and guide for using the Qiskit.

How to Install Qiskit – Programming on Quantum Computers Season 1 Ep 2

Getting started with Qiskit

Anaconda promt ⇒ pip install qiskit

```
(base) C:¥Users¥admin>pip install qiskit
Requirement already satisfied: qiskit in c:¥users¥admin¥anaconda3¥lib¥site-packages (0.35.0)
Requirement already satisfied: qiskit-ignis==0.7.0 in c:¥users¥admin¥anaconda3¥lib¥site-packages (from qiskit) (0.7.0)
Requirement already satisfied: qiskit-aer==0.10.3 in c:¥users¥admin¥anaconda3¥lib¥site-packages (from qiskit) (0.10.3)
Requirement already satisfied: qiskit-terra==0.20.0 in c:¥users¥admin¥anaconda3¥lib¥site-packages (from qiskit) (0.20.0)

Requirement already satisfied: qiskit-ibmq-provider==0.18.3 in c:¥users¥admin¥anaconda3¥lib¥site-packages (from qiskit)
(0.18.3)
Requirement already satisfied: numpy>=1.16.3 in c:¥users¥admin¥anaconda3¥lib¥site-packages (from qiskit-aer==0.10.3->qis
kit) (1.20.3)
Requirement already satisfied: scipy>=1.0 in c:¥users¥admin¥anaconda3¥lib¥site-packages (from qiskit-aer==0.10.3->qiskit
) (1.7.1)
Requirement already satisfied: requests-ntlm>=1.1.0 in c:¥users¥admin¥anaconda3¥lib¥site-packages (from qiskit-ibmq-prov
ider==0.18.3->qiskit) (1.1.0)
Requirement already satisfied: urllib3>=1.21.1 in c:¥users¥admin¥anaconda3¥lib¥site-packages (from qiskit-ibmq-provider=
=0.18.3->qiskit) (1.26.7)
Requirement already satisfied: requests>=2.19 in c:¥users¥admin¥anaconda3¥lib¥site-packages (from qiskit-ibmq-provider==
0.18.3->qiskit) (2.26.0)
Requirement already satisfied: python-dateutil>=2.8.0 in c:¥users¥admin¥anaconda3¥lib¥site-packages (from qiskit-ibmq-pr
ovider==0.18.3->qiskit) (2.8.2)
Requirement already satisfied: websocket-client>=1.0.1 in c:¥users¥admin¥anaconda3¥lib¥site-packages (from qiskit-ibmq-p
rovider==0.18.3->qiskit) (1.3.2)
Requirement already satisfied: retworkx>=0.8.0 in c:¥users¥admin¥anaconda3¥lib¥site-packages (from qiskit-ignis==0.7.0->
```

Getting started with Qiskit

In [31]:
```
import qiskit
import matplotlib.pyplot
from math import pi
from qiskit.visualization import plot_bloch_vector, plot_bloch_multivector
```

In [2]:
```
qiskit.__qiskit_version__
```

```
{'qiskit-terra': '0.20.0', 'qiskit-aer': '0.10.3', 'qiskit-ignis': '0.7.0', 'qiskit-ibmq-provider': '0.18.3', 'qiskit-aqua': None,
'qiskit': '0.35.0', 'qiskit-nature': None, 'qiskit-finance': None, 'qiskit-optimization': None, 'qiskit-machine-learning': None}
```

In [3]:
```
from qiskit import IBMQ
```

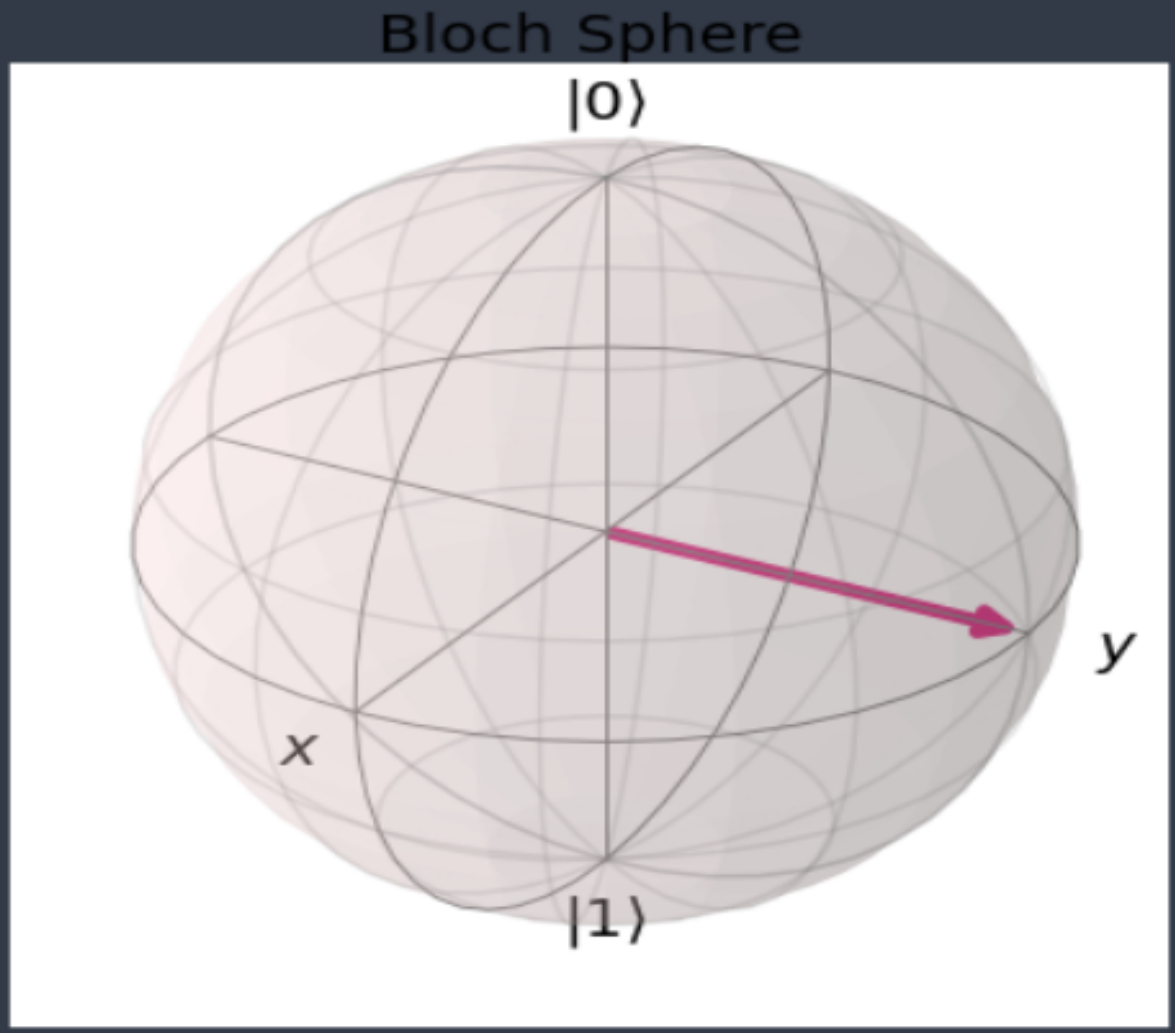
Getting started with Qiskit

```
from qiskit import IBMQ

IBMQ.save_account(
```


```

IBMQ.load_account()
```

Drawing Bloch sphere using Qiskit

```
plot_bloch_vector([0,1,0], title = "Bloch Sphere")
```



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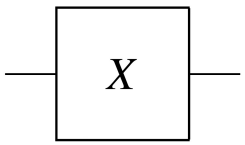
Quantum Circuits : A review

Definition

A quantum gate is a unitary transformation $\mathcal{U} : \mathbb{H}^{\otimes n} \to \mathbb{H}^{\otimes n}$.
If $n = 1$, we call it unary, and if $n = 2$, we call it binary.

Three representations of a quantum gate: Operator, matrix, and symbol.

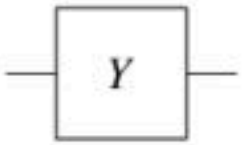
Examples of unary transformation : $V : \mathbb{H} \to \mathbb{H}$, unitary 2×2 matrix.
1. Pauli-X gate (Or Q-Not gate)



$$X := \sigma_x \qquad \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

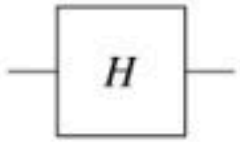
Quantum Circuits : A review

2. Pauli-Y gate



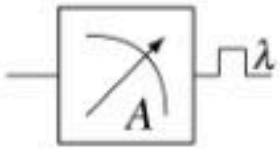
$$Y := \sigma_y \qquad \begin{pmatrix} 0 & -\mathrm{i} \\ \mathrm{i} & 0 \end{pmatrix}$$

3. Hadamard gate



$$H := \frac{\sigma_x + \sigma_z}{\sqrt{2}} \qquad \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

4, Measurement of observable A (this is not a gate).



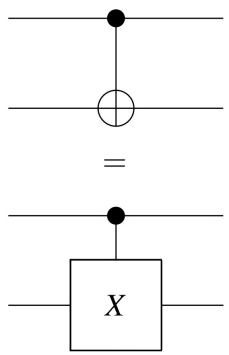
[Not a gate, but a non-unitary transformation of the input-state to an eigenstate of A and delivery of measured value λ]

Quantum Circuits : A review

Examples of binary quantum gate : $\mathcal{U} : \mathbb{H}^{\otimes 2} \to \mathbb{H}^{\otimes 2}$ with basis $\{|0\rangle^2, |1\rangle^2, |2\rangle^2, |3\rangle^2\}$.

1. $\Lambda^1(X)$

Controlled NOT (C-NOT)

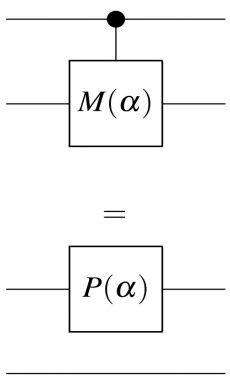


$$\Lambda^1(X) = |0\rangle\langle 0| \otimes \mathbf{1} + |1\rangle\langle 1| \otimes X$$

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

2.

Controlled phase-multiplication



$$\Lambda^1(M(\alpha)) = |0\rangle\langle 0| \otimes \mathbf{1} + |1\rangle\langle 1| \otimes e^{i\alpha}\mathbf{1} = P(\alpha) \otimes \mathbf{1}$$

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & e^{i\alpha} & 0 \\ 0 & 0 & 0 & e^{i\alpha} \end{pmatrix}$$

Quantum Circuits : A review

Definition

Let $n, L \in \mathbb{N}$ and let $\mathcal{U}_1, \cdots, \mathcal{U}_L \in \mathcal{U}(\mathbb{H}^{\otimes n})$ be quantum n-gates. Then the composition operator $U = \mathcal{U}_L \cdots \mathcal{U}_1$ is called the quantum circuit of depth L.

Remark : There are two other definitions of the quantum circuit. See our previous seminar book on the quantum computation.

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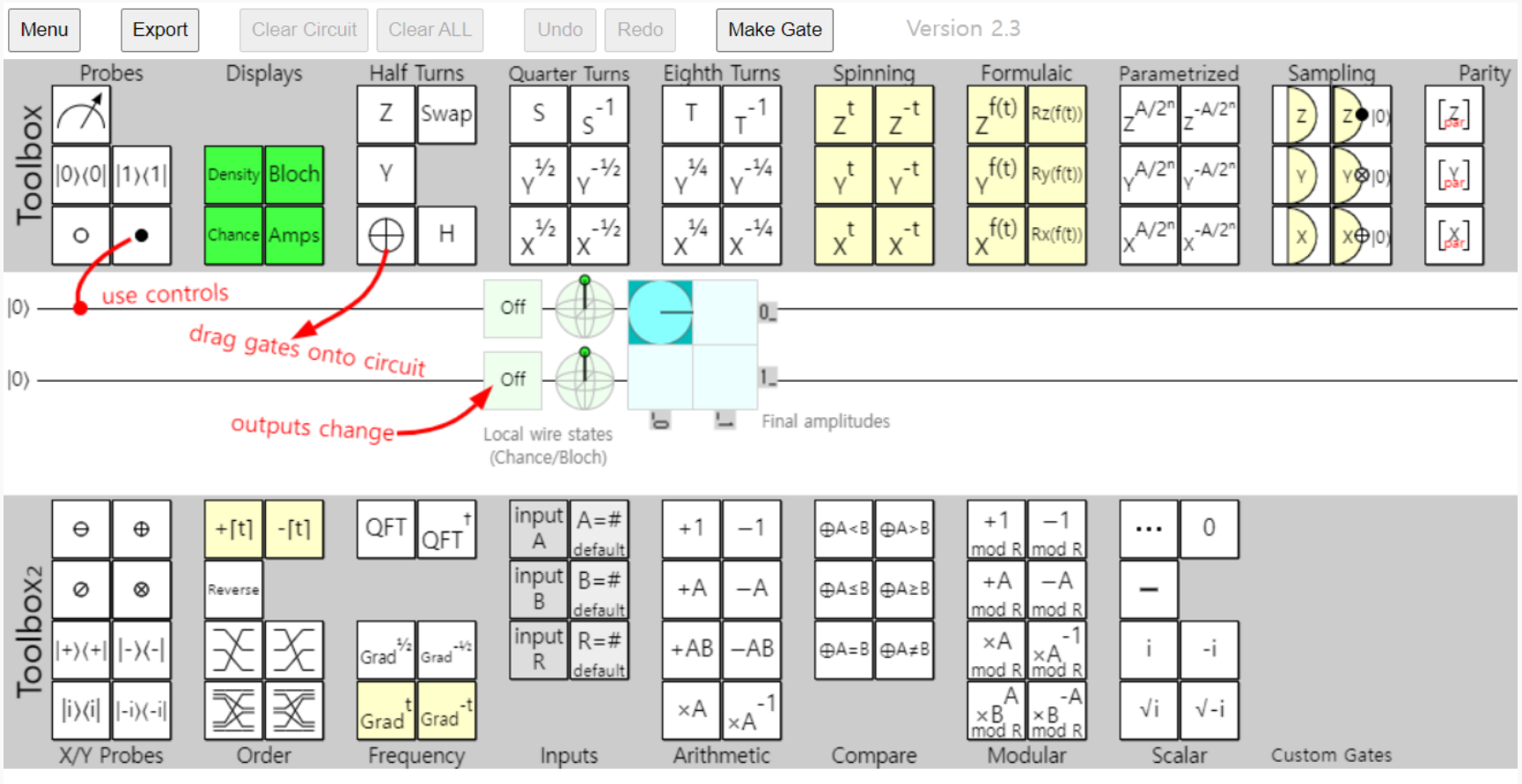
However, before those... let's play with

https://algassert.com/quirk



Getting started with Cirq

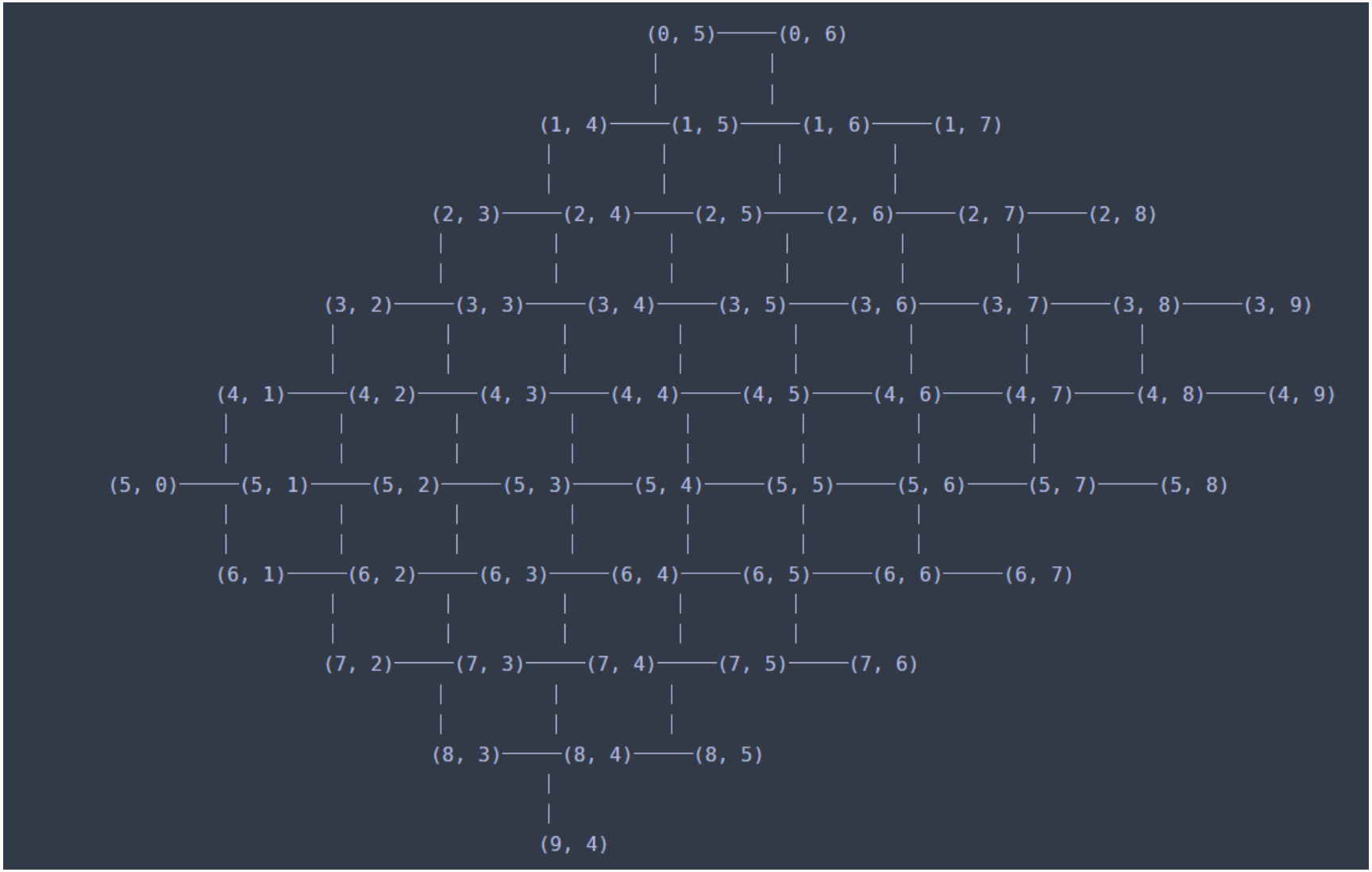
Please see : https://www.qmunity.tech/tutorials/introduction-to-cirq

```
In [ ]: pip install cirq
```

```
In [1]: import cirq
```

```
In [2]: import cirq
        import cirq_google
```

```
In [3]: print(cirq_google.Sycamore)   #Not Bristlecone it is deprecated
```


(0, 5)——(0, 6)
(1, 4)——(1, 5)——(1, 6)——(1, 7)
(2, 3)——(2, 4)——(2, 5)——(2, 6)——(2, 7)——(2, 8)
(3, 2)——(3, 3)——(3, 4)——(3, 5)——(3, 6)——(3, 7)——(3, 8)——(3, 9)
(4, 1)——(4, 2)——(4, 3)——(4, 4)——(4, 5)——(4, 6)——(4, 7)——(4, 8)——(4, 9)
(5, 0)——(5, 1)——(5, 2)——(5, 3)——(5, 4)——(5, 5)——(5, 6)——(5, 7)——(5, 8)
(6, 1)——(6, 2)——(6, 3)——(6, 4)——(6, 5)——(6, 6)——(6, 7)
(7, 2)——(7, 3)——(7, 4)——(7, 5)——(7, 6)
(8, 3)——(8, 4)——(8, 5)
(9, 4)

Creating qubits

In [12]:

```
q0=cirq.NamedQubit("q0")
q1=cirq.NamedQubit("q1")
print(q0, q1)
# Line qubits
q3 = cirq.LineQubit(4)
print(q3)
# Creates LineQubit(0), LineQubit(1), LineQubit(2)
q0, q1, q2 = cirq.LineQubit.range(3)
print(q0,q1,q2)
# Grid Qubits can also be referenced individually
q0_1= cirq.GridQubit(0,1)
print(q0_1)
# Create 16 qubits from (0,0) to (7,7)
qubits = cirq.GridQubit.square(8)
print(qubits)
```

Creating qubits : Result

```
q0 q1
4
0 1 2
(0, 1)
[cirq.GridQubit(0, 0), cirq.GridQubit(0, 1), cirq.GridQubit(0, 2), cirq.GridQubit(0, 3), cirq.GridQubit(0, 4), cirq.GridQubit(0, 5),
cirq.GridQubit(0, 6), cirq.GridQubit(0, 7), cirq.GridQubit(1, 0), cirq.GridQubit(1, 1), cirq.GridQubit(1, 2), cirq.GridQubit(1, 3),
cirq.GridQubit(1, 4), cirq.GridQubit(1, 5), cirq.GridQubit(1, 6), cirq.GridQubit(1, 7), cirq.GridQubit(2, 0), cirq.GridQubit(2, 1),
cirq.GridQubit(2, 2), cirq.GridQubit(2, 3), cirq.GridQubit(2, 4), cirq.GridQubit(2, 5), cirq.GridQubit(2, 6), cirq.GridQubit(2, 7),
cirq.GridQubit(3, 0), cirq.GridQubit(3, 1), cirq.GridQubit(3, 2), cirq.GridQubit(3, 3), cirq.GridQubit(3, 4), cirq.GridQubit(3, 5),
cirq.GridQubit(3, 6), cirq.GridQubit(3, 7), cirq.GridQubit(4, 0), cirq.GridQubit(4, 1), cirq.GridQubit(4, 2), cirq.GridQubit(4, 3),
cirq.GridQubit(4, 4), cirq.GridQubit(4, 5), cirq.GridQubit(4, 6), cirq.GridQubit(4, 7), cirq.GridQubit(5, 0), cirq.GridQubit(5, 1),
cirq.GridQubit(5, 2), cirq.GridQubit(5, 3), cirq.GridQubit(5, 4), cirq.GridQubit(5, 5), cirq.GridQubit(5, 6), cirq.GridQubit(5, 7),
cirq.GridQubit(6, 0), cirq.GridQubit(6, 1), cirq.GridQubit(6, 2), cirq.GridQubit(6, 3), cirq.GridQubit(6, 4), cirq.GridQubit(6, 5),
cirq.GridQubit(6, 6), cirq.GridQubit(6, 7), cirq.GridQubit(7, 0), cirq.GridQubit(7, 1), cirq.GridQubit(7, 2), cirq.GridQubit(7, 3),
cirq.GridQubit(7, 4), cirq.GridQubit(7, 5), cirq.GridQubit(7, 6), cirq.GridQubit(7, 7)]
```

Creating unitary matrices

```
In [5]: print(cirq.unitary(cirq.X)) #Unitary of the X gate
```

```
[[0.+0.j 1.+0.j]
 [1.+0.j 0.+0.j]]
```

```
In [6]: a, b = cirq.LineQubit.range(2) #Unitary of SWAP operator on two qubits
        print(cirq.unitary(cirq.SWAP(a, b)))
```

```
[[1.+0.j 0.+0.j 0.+0.j 0.+0.j]
 [0.+0.j 0.+0.j 1.+0.j 0.+0.j]
 [0.+0.j 1.+0.j 0.+0.j 0.+0.j]
 [0.+0.j 0.+0.j 0.+0.j 1.+0.j]]
```

Creating Circuits

In [49]:
```
circuit=cirq.Circuit(cirq.H(q0), cirq.CNOT(q0,q1), cirq.measure(q0,q1))

print(circuit)
```

```
0: ———H———@———M———
          |   |
1: ———————X———M———
```

In [53]:
```
simulator=cirq.Simulator()

result=simulator.run(circuit,repetitions=30)
print(result)
```

```
0,1=10010110100111111011001111011, 10010110100111111011001111011
```

Creating the Bell state

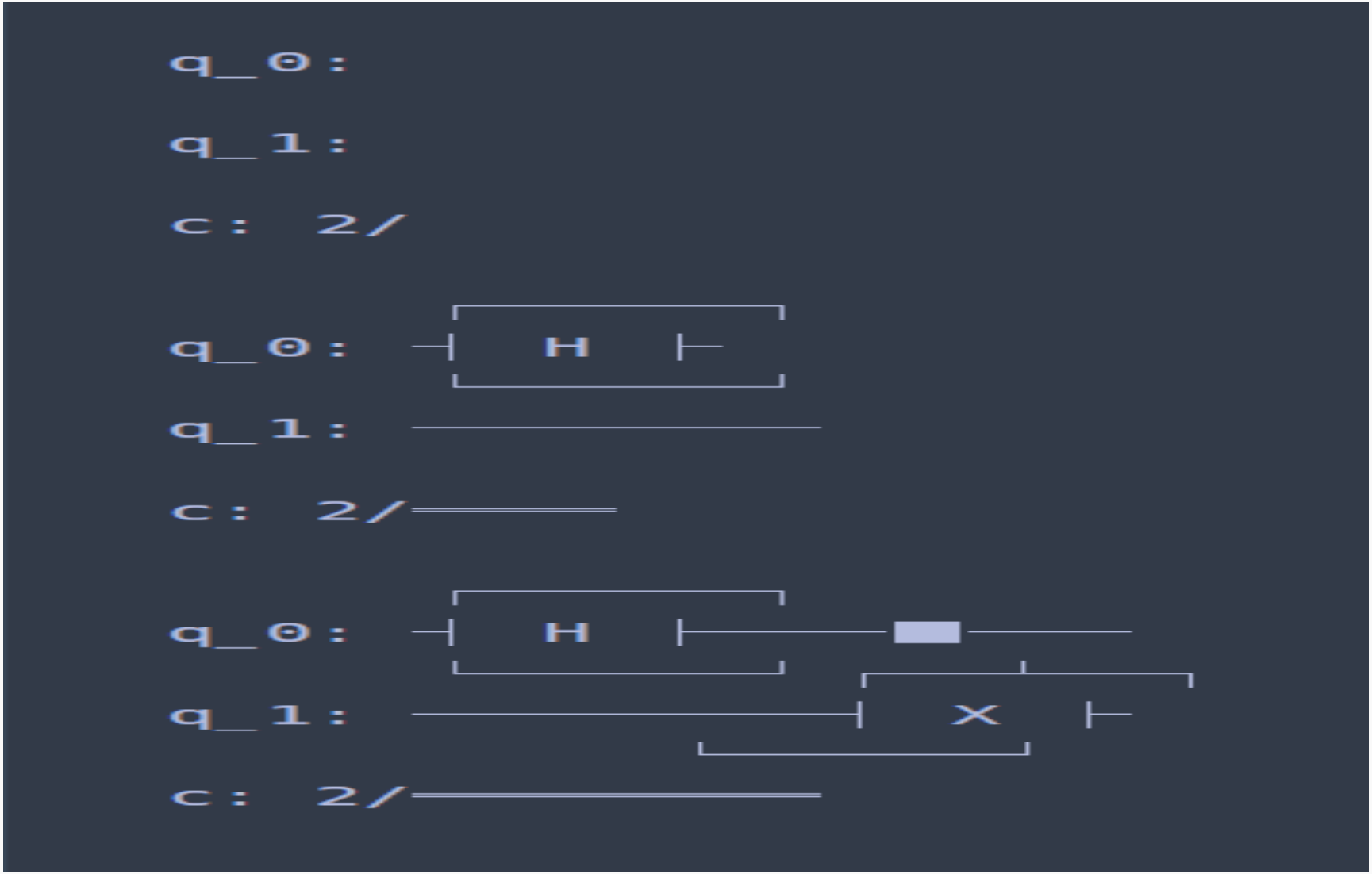
```
# We will generate the Bell State:
# sqrt(2) * ( |00> + |11> )
bell_circuit = cirq.Circuit()
a, b = cirq.LineQubit.range(2)
bell_circuit.append(cirq.H(a))
bell_circuit.append(cirq.CNOT(a,b))
print(bell_circuit)
# Initialize Simulator
s=cirq.Simulator()

print('Simulate the circuit:')
results=s.simulate(bell_circuit)
print(results)
print()

# For sampling, we need to add a measurement at the end
bell_circuit.append(cirq.measure(a, b, key='result'))

print('Sample the circuit:')
samples=s.run(bell_circuit, repetitions=1000)
```

Creating the Bell state : The result

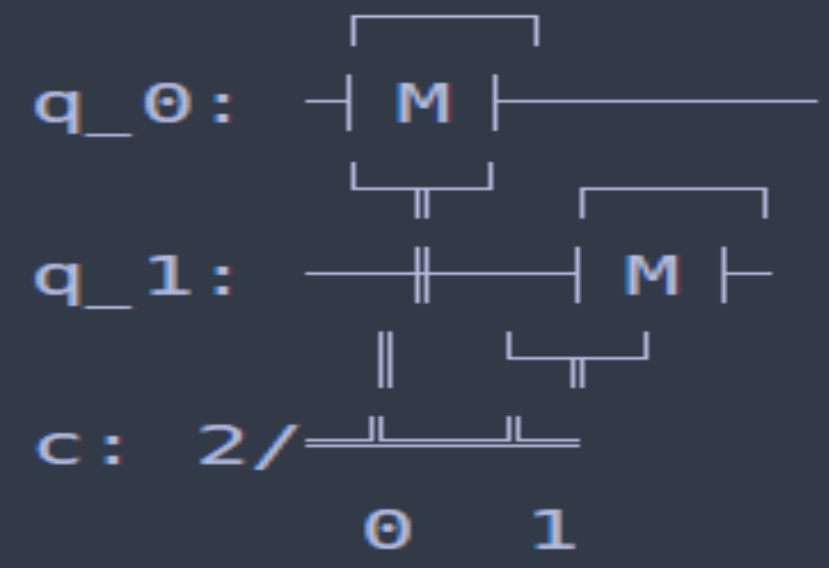
```
0: ──────H──────@──────
                │
1: ─────────────X──────
Simulate the circuit:
measurements: (no measurements)

qubits: (cirq.LineQubit(0), cirq.LineQubit(1))
output vector: 0.707|00⟩ + 0.707|11⟩

phase:
output vector: |⟩

Sample the circuit:
```

Qiskit again, Bell state again

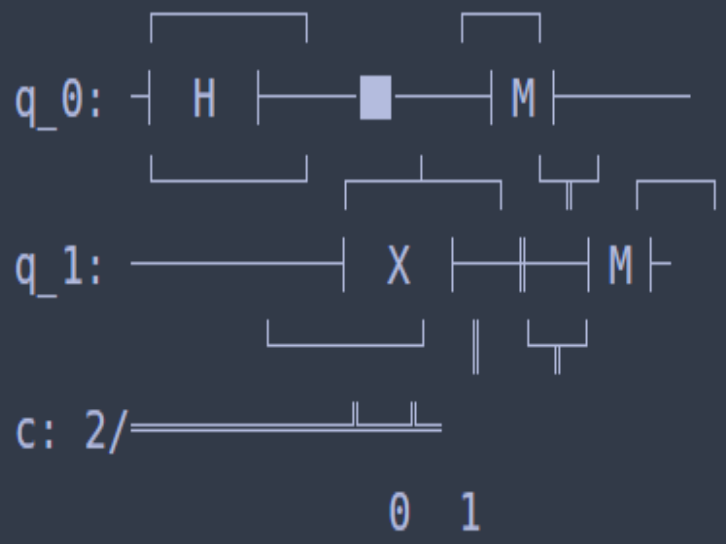
```
#quantum circuit to create a Bell state
bell = QuantumCircuit(2,2)
print(bell)
bell.h(0)
print(bell)
bell.cx(0,1)
print(bell)
```


q_0:
q_1:
c: 2/
q_0:
H
q_1:
c: 2/
q_0:
H
q_1:
X
c: 2/

```
meas = QuantumCircuit(2,2)
meas.measure([0,1],[0,1])
print(meas)
```



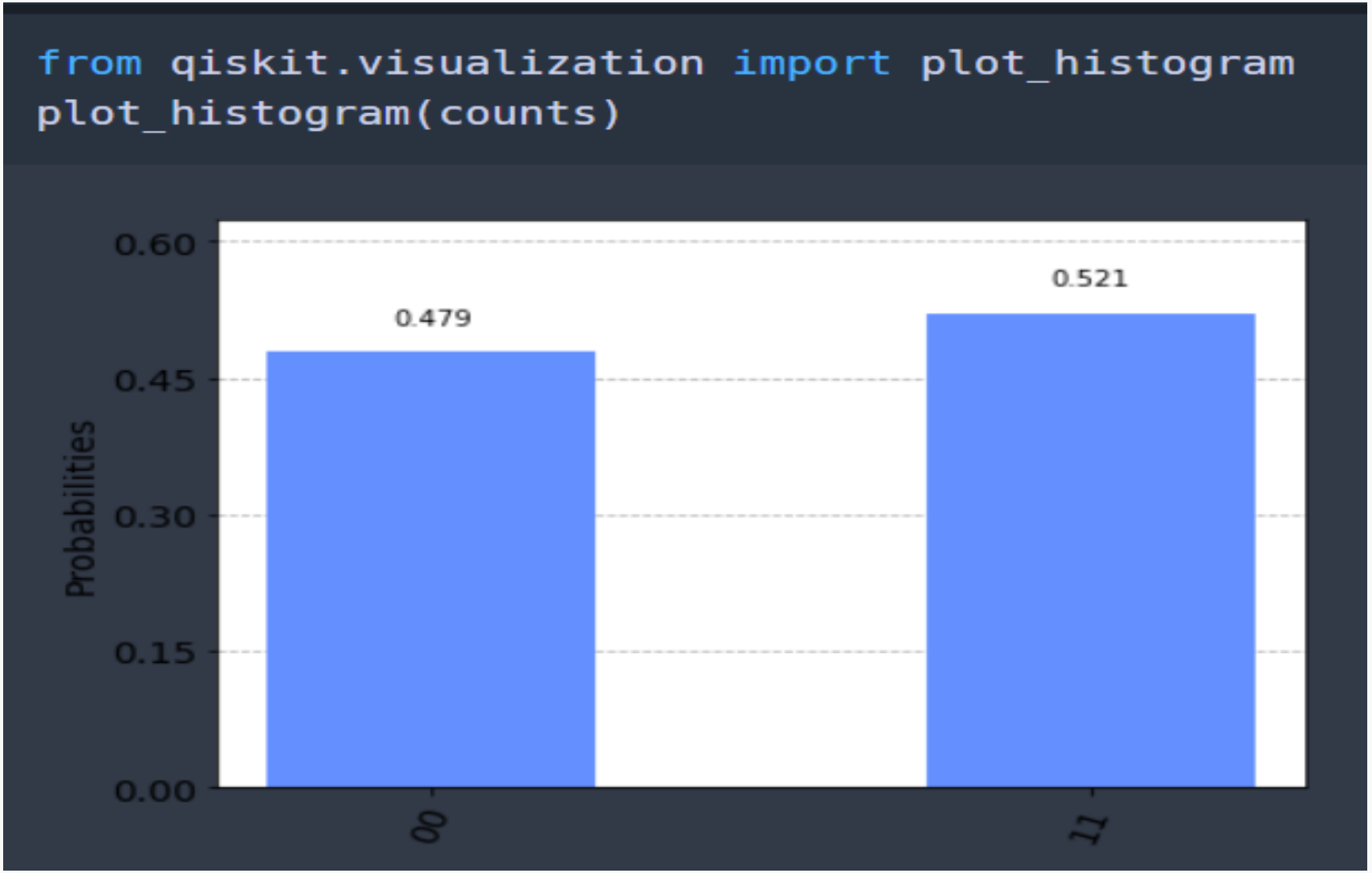
```
#execute the quantum circuit
backend = BasicAer.get_backend('qasm_simulator') #the device the run on
circ = bell + meas
print(circ)
```



```
result = execute(circ, backend, shots = 1000).result()
counts = result.get_counts(circ)
print(counts)
```

```
{'11': 521, '00': 479}
```

```
from qiskit.visualization import plot_histogram
plot_histogram(counts)
```



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Von Neumann Entropy

Quantum counterpart of the Shannon entropy

Definition

Given a mixed state $\rho = \sum_{i=1}^{n} p_i |\psi_i\rangle\langle\psi_i|$, the von Neumann entropy is given by

$$S(\rho) = -Tr(\rho \log \rho)$$

It is...

1. subadditive, $S(\rho_{12}) \leq S(\rho_1) + S(\rho_2)$.
2. $0 \leq S \leq N = dim(\mathbb{H})$.

Using Qiskit, one can compute the evolution of the von Neumann entropy of a quantum state.

see you next week :)